

Genesis_Medicine - 17_r16_topical_chromanol_lead

R16 Topical Chromanol Lead Short Communication: 18-Pair 30 ns Matrix, 60 ns Robustness Panels, and Complete 200 ns Anchor-Triad Follow-up

Abstract

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Abstract

R16 optimized the R15 chromanol fragment toward a topical lead hypothesis by increasing skin-window compatibility while preserving a compact chromanol core. We evaluated six chloro/dimethyl analogs across TGFB1, DCT, and TYR using Boltz-2 cofolding, an 18-pair 30 ns OpenMM stability matrix, two 60 ns robustness panels, and 100 ns plus 200 ns anchor-triad follow-up. The top cofold row was r16_03_tgfb1 (R15_chromanol_CL_pos9, 0CC1C0c2cc(0)c(CL)cc2C1) with affinity probability 0.682. All 18 30 ns matrix entries were stable, the TGFB1 top-six 60 ns panel completed 6/6 stable, and the DCT/TYR representative 60 ns panel completed 3/3 stable. The 200 ns long-horizon anchor triad also completed 3/3 stable: TGFB1 R15_chromanol_CL_pos9 max RMSD 0.71 Å, DCT R15_chromanol_CL_pos9 max RMSD 1.05 Å, and TYR

R15_chromanol_Cl_pos6 max RMSD 0.80 Å. These data support an in-silico topical optimization hypothesis, not clinical efficacy, confirmed binding, composition novelty, or freedom to operate.

Keywords: chromanol, topical drug discovery, TGFB1, DCT, TYR, Boltz-2, OpenMM, in silico, prior-art gate

1. Research Question

This manuscript separates the topical R16 analog path from the R15 safety-first parent path. The question is whether R16 chloro/dimethyl chromanol analogs can be prioritized as topical in-silico candidates after target cofolding, raw-pose sanity checks, 30-60 ns robustness panels, and complete 200 ns long-horizon anchor follow-up.

2. Data Sources

file	role
pilot/cpu_meaningful/ r16_chromanol_topical_manifest.csv	six R16 topical chromanol analog definitions and topical scores
pilot/cpu_meaningful/ r16_chromanol_topical_cofold.csv	18 Boltz-2 cofold rows across TGFB1, DCT, and TYR
pilot/cpu_meaningful/ r16_topical_chromanol_30ns_matrix_summary.csv	18-pair 30 ns MD matrix
pilot/ md_r16_chromanol_topical_tgfb1_to_p6_60ns/summary.json	TGFB1 top-six 60 ns robustness panel
pilot/ md_r16_chromanol_topical_pigment_representative_60ns/summary.json	DCT/TYR representative 60 ns panel
pilot/ md_r16_chromanol_anchor_triad_100ns/summary.json	TGFB1/DCT/TYR 100 ns anchor triad
pilot/ md_r16_chromanol_anchor_triad_200ns/summary.json	complete 200 ns long-horizon anchor triad
pilot/cpu_meaningful/ precompute_prior_art_gate.csv	technical prior-art pre-gate, not legal FTO opinion

3. Results

3.1 Analog properties

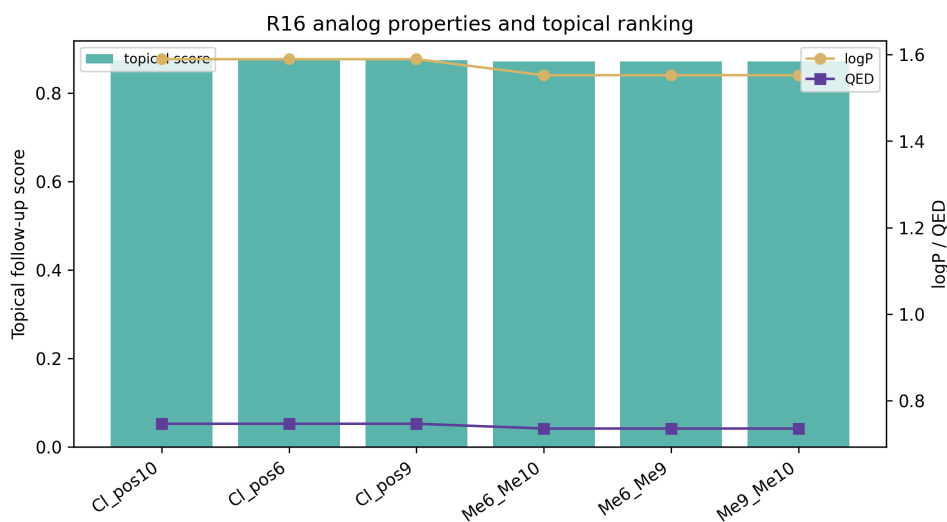


Figure 1. R16 analog topical scores with logP and QED overlays.

3.2 Cofold ranking

rank	job_id	target	analog	affinity probability	logP	QED
1	r16_03_tgfb1	TGFB1	R15_chromanol_Cl_pos9	0.682	1.59	0.747
2	r16_05_tgfb1	TGFB1	R15_chromanol_Me6_Me9	0.636	1.55	0.736
3	r16_04_tgfb1	TGFB1	R15_chromanol_Me6_Me10	0.616	1.55	0.736
4	r16_06_tgfb1	TGFB1	R15_chromanol_Me9_Me10	0.615	1.55	0.736
5	r16_02_tgfb1	TGFB1	R15_chromanol_Cl_pos6	0.612	1.59	0.747
6	r16_01_tgfb1	TGFB1	R15_chromanol_Cl_pos10	0.587	1.59	0.747
7	r16_03_dct	DCT	R15_chromanol_Cl_pos9	0.546	1.59	0.747
8	r16_02_dct	DCT	R15_chromanol_Cl_pos6	0.536	1.59	0.747
9	r16_02_tyrr	TYR	R15_chromanol_Cl_pos6	0.525	1.59	0.747
10		DCT		0.502	1.59	0.747

rank	job_id	target	analog	affinity probability	logP	QED
	r16_01_dct		R15_chromanol_Cl_pos10			

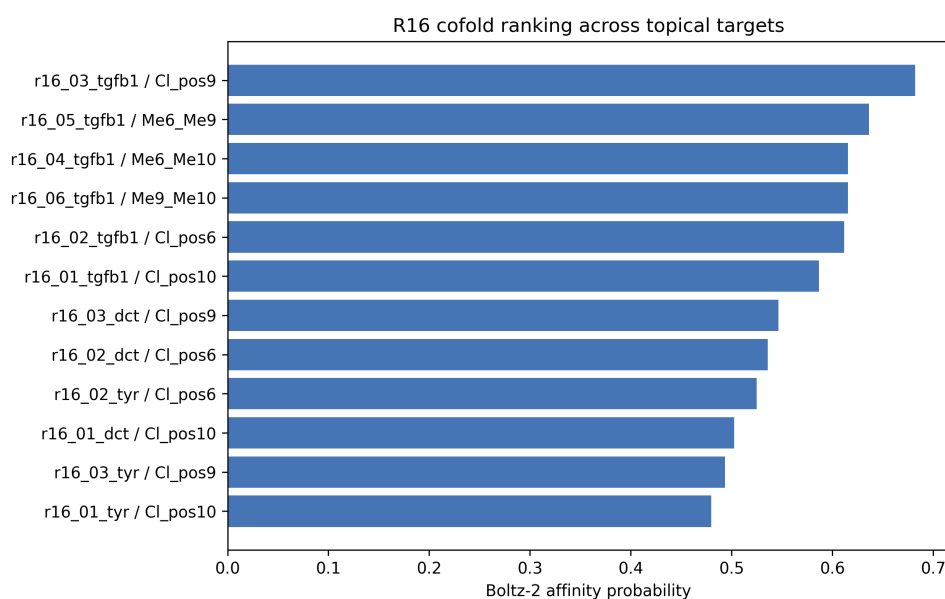


Figure 5. R16 cofold ranking across TGFB1, DCT, and TYR target pairs.

3.3 Complete 18-pair 30 ns matrix

All 18 R16 chromanol topical target pairs completed 30 ns MD with `stable_30ns=True`. Across the matrix, the maximum RMSD was 1.38 Å and the maximum last-third RMSD was 1.17 Å.

job_id	target	analog	affinity probability	mean RMSD Å	last-third RMSD Å	max RMSD Å
r16_06_dct	DCT	R15_chromanol_Me9_Me10	0.420	0.22	0.18	0.68
r16_05_dct	DCT	R15_chromanol_Me6_Me9	0.421	0.45	0.43	0.81
r16_04_dct	DCT	R15_chromanol_Me6_Me10	0.329	0.51	0.53	0.68
r16_02_dct	DCT	R15_chromanol_Cl_pos6	0.536	0.56	0.59	0.81
r16_03_dct	DCT	R15_chromanol_Cl_pos9	0.546	0.38	0.67	1.00
r16_01_dct	DCT	R15_chromanol_Cl_pos10	0.502	0.74	1.00	1.25
r16_01_tgfb1	TGFB1		0.587	0.21	0.19	0.66

job_id	target	analog	affinity probabili ty	mean RMSD A	last- third RMSD A	max RMSD A
		R15_chro manol_Cl_ pos10				
r16_06_tg fb1	TGFB1	R15_chro manol_Me 9_Me10	0.615	0.54	0.53	0.81
r16_02_tg fb1	TGFB1	R15_chro manol_Cl_ pos6	0.612	0.55	0.67	0.87
r16_05_tg fb1	TGFB1	R15_chro manol_Me 6_Me9	0.636	0.68	0.71	0.89
r16_04_tg fb1	TGFB1	R15_chro manol_Me 6_Me10	0.616	0.99	1.04	1.24
r16_03_tg fb1	TGFB1	R15_chro manol_Cl_ pos9	0.682	0.81	1.17	1.38
r16_01_ty r	TYR	R15_chro manol_Cl_ pos10	0.480	0.28	0.15	0.68
r16_03_ty r	TYR	R15_chro manol_Cl_ pos9	0.493	0.24	0.21	0.67
r16_05_ty r	TYR	R15_chro manol_Me 6_Me9	0.422	0.46	0.27	0.93
r16_06_ty r	TYR	R15_chro manol_Me 9_Me10	0.430	0.46	0.34	0.71
r16_04_ty r	TYR	R15_chro manol_Me 6_Me10	0.351	0.36	0.59	0.74
r16_02_ty r	TYR	R15_chro manol_Cl_ pos6	0.525	0.69	0.72	0.91



Figure 2. R16 18-pair 30 ns matrix. Values are last-third ligand RMSD.

3.4 60 ns robustness panels

The TGFB1 top-six 60 ns panel completed 6/6 stable entries, and the representative DCT/TYR pigmentation panel completed 3/3 stable entries. The strongest single analog across both target families is R15_chromanol_Cl_pos9, which carries TGFB1 and DCT support. R15_chromanol_Cl_pos6 is the strongest TYR-focused follow-up.

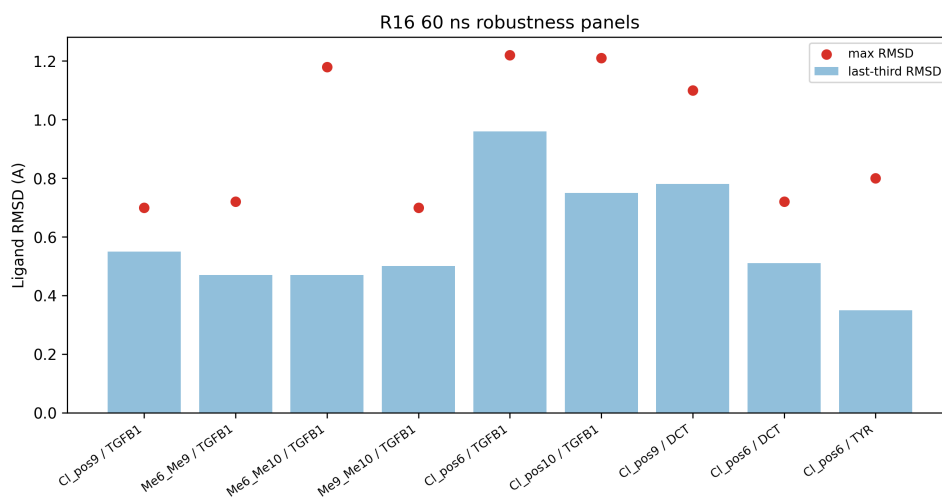


Figure 3. R16 60 ns robustness panels for TGFB1 and representative pigmentation targets.

3.5 Complete 100 ns and 200 ns anchor triads

name	target	analog	affinity probability	mean RMSD Å	last-third RMSD Å	max RMSD Å
r16_03_tgfb1_R15_chromanol_Cl_pos9_200ns	TGFB1	R15_chromanol_Cl_pos9	0.682	0.27	0.24	0.71

name	target	analog	affinity probability	mean RMSD A	last-third RMSD A	max RMSD A
r16_03_dct_R15_chromanol_Cl_pos9_200ns	DCT	R15_chromanol_Cl_pos9	0.546	0.34	0.43	1.05
r16_02_tyr_R15_chromanol_Cl_pos6_200ns	TYR	R15_chromanol_Cl_pos6	0.525	0.49	0.57	0.80

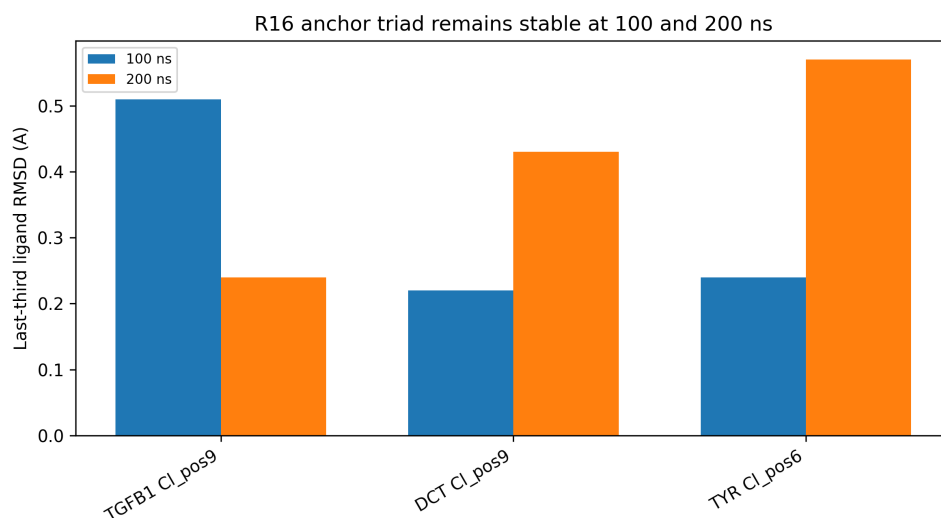


Figure 4. R16 anchor triad last-third RMSD at 100 and 200 ns.

3.6 Pose sanity and prior-art gates

The R16 raw-pose sanity split is 11 pass, 7 review, and 0 fail rows. Review-level rows are disclosed as raw Boltz-pose caveats and interpreted with minimized/MD stability.

The precompute prior-art gate classifies the R16 chromanol analog rows as {'hold_expensive_compute_until_markush_review': 9, 'cheap_compute_allowed_prior_art_pending': 9}. The practical implication is strict: existing R16 data can be written as in-silico prioritization, but new 100-200 ns expansion, RBE/ABFE, synthesis/purchase, commercial novelty, and FTO claims should wait for PubChem/SureChEMBL/PATENTSCOPE/Lens/EPO OPS plus professional Markush and attorney claim-chart review.

4. Development Interpretation

R15_chromanol_Cl_pos9 is the strongest TGFB1-first topical hypothesis because it combines the highest cofold probability, stable 30 ns target-diverse MD, stable TGFB1 60 ns MD, representative DCT 60 ns support, and stable TGFB1/DCT 200 ns long-horizon anchors.

R15_chromanol_Cl_pos6 is the strongest TYR-focused pigmentation follow-up because it carries DCT/TYR 60 ns support and stable TYR 200 ns anchoring. This is a prioritization statement only.

5. Limitations

All findings are in silico only. Boltz-2 cofolding is not a biochemical assay. MD pose stability is not potency, target engagement, residence time, or skin exposure. Photosafety, sensitization, hERG, AMES, DILI, irritation, IVRT/IVPT, PBPK, formulation compatibility, and target-engagement assays remain required. A PubChem no_hit or local distinctness result is not a freedom-to-operate opinion because Markush and use claims can still cover unmade or unlisted analogs.

6. Conclusion

R16 is the strongest current topical chromanol in-silico lead family in Genesis_Medicine. The completed 30 ns, 60 ns, 100 ns, and 200 ns panels justify manuscript completion and CRO hypothesis packaging, while the prior-art gate blocks stronger commercial and follow-on expensive-compute claims until Markush/FTO review is complete.